

# Dai Tonghua

Algorithm Engineer · Algorithm–Hardware Co-Design · PhD Candidate

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## CORE AREAS

- Core capability: **dual-stack algorithm × hardware co-design** — algorithm side covers model, optimization and quantization algorithms; implementation side covers FPGA / ASIC and RTL, with on-chip deployment cost as the ultimate optimization target.
- Quantization algorithms**: end-to-end model / signal quantization — mixed-precision quantization, quantization-aware training (QAT), improved quantization operators, gradient and iterative optimizer-state compression; unified bit-width optimization across parameters, signal path, storage, gradients and optimizer, **losslessly down to X bit** versus the engineering baseline.
- Chip design**: PPA-driven algorithm–hardware co-design, with scheme trade-offs forced by ASIC resource limits; lossless low-cost low-power schemes based on approximate computing and approximate operators; contributed to RTL implementation and to the FPGA (+ ROC) algorithm verification platform, spanning the algorithm → RTL → FPGA chain. RF nonlinear-compensation (DPD) model quantization lands under exactly these constraints.
- Wireless sensing algorithms**: defined and solved the generalized Wi-Fi cross-domain sensing problem, including single-link zero-effort cross-domain recognition, open-set gesture detection and few-shot open-set recognition.
- Mathematical foundation**: MIPT Applied Math & Physics, GPA 5.0/5.0; B.Sc. in applied mathematics and physics, passed the Landau's Theoretical Minimum in theoretical physics, with an SCI Q2 paper (*Ultrasonics*); algorithms derived from first principles.

## SKILLS

|             |                      |                              |                                         |                                |                  |
|-------------|----------------------|------------------------------|-----------------------------------------|--------------------------------|------------------|
| Algorithms  | Neural networks      | NN quantization              | DPD / nonlinear compensation            | Communication algorithms / DSP | Wireless sensing |
|             |                      | Few-shot / open-set learning |                                         |                                |                  |
| Hardware    | RTL design / Verilog | FPGA / ASIC                  | Approximate computing / fused operators |                                |                  |
| Engineering | Python               | MATLAB                       | C++                                     |                                |                  |
| Languages   | Chinese (Native)     | Russian (Fluent)             | English (Professional)                  |                                |                  |

## PROFESSIONAL EXPERIENCE

### Algorithm Engineer — Huawei Technologies, Moscow LRI

Oct 2021 – Present

#### Quantization & algorithm–hardware co-design (Fellow team)

- The RF nonlinear-compensation model (online-training-like NN) is resource-constrained on ASIC. Designed a quantization scheme covering parameters, signal path, storage and iterative optimizer; developed a whole-model quantization framework for the full RF front-end chain — mixed-precision quantization, QAT, advanced quantization operators, with PPA-driven scheme design; versus the engineering baseline this is **lossless down to 4–12 bit**, with **chip resources down over 30% per generation** at equal performance.
- Implemented unified bit-width optimization across parameters, signal path, storage and optimizer for the online-training model, keeping convergence fast at low bit-width.
- Owned lossless low-cost, low-power schemes based on approximate computing and approximate operators; led ASIC frontier exploration with local academic collaborators.
- Contributed to version improvements and debugging of the FPGA + ROC algorithm verification platform, spanning the algorithm → RTL → FPGA verification chain.

**Awards**: Golden Seed Award · 2022 Annual Best Support · Continuous Contribution

### Invited Lecturer — HSE University

Jan 2025 – Jul 2025

<https://www.hse.ru/en/edu/courses/904327944>

- Sole lecturer of *Systems and Networks of Information Transfer* (principles of communication networks and basic data-transfer protocols), taught bilingually in Russian and English.

## RESEARCH

### DPD Coefficient Quantization — M.Sc. Thesis — MIPT

2021 – 2023

- Developed quantization methods for digital predistortion (DPD) coefficients, enabling low-power algorithm implementation for communication RF digital chips; together with industry RF nonlinear-model quantization this forms a complete DPD algorithm-to-hardware chain.

### Wi-Fi Generalized Cross-Domain Gesture Recognition — Ph.D. — HSE University

2023 – 2027 (expected)

- Dissertation: methods of Wi-Fi-based gesture recognition under generalized cross-domain tasks. Independently framed the proposal and formally generalized the Wi-Fi cross-domain sensing problem, establishing a new research direction.
- First Wi-Fi-based **single-link, zero-effort cross-domain** gesture recognition system.
- First **few-shot open-set recognition** framework for Wi-Fi gesture recognition.
- First **out-of-distribution detection** framework for Wi-Fi gestures robust to semantic and covariate shift.
- First few-shot Wi-Fi gesture recognition method addressing **support–query shift** under dual cross-domain settings.

## EDUCATION

### Ph.D. — HSE University

2023 – 2027 (expected)

Communication Systems, Networks and Devices

Thesis: Methods of Wi-Fi-based gesture recognition in generalized cross-domain tasks

Advisor: Evgeny Khorov (Head of Wireless Networks Lab at IITP RAS)

Russian Government Scholarship

### M.Sc. with Distinction — Moscow Institute of Physics and Technology (MIPT)

2021 – 2023

Applied Mathematics and Physics · GPA 5.0/5.0

Thesis: Development of quantization methods for digital predistortion coefficients

MIPT–Huawei joint program (work completed within the IRF Lab framework; advisor — internal expert)

Russian Government Scholarship · China Scholarship Council (CSC)

### B.Sc. — Moscow Institute of Physics and Technology (MIPT)

2016 – 2021

Applied Mathematics and Physics; research area: solid-state physics (magnetoelastic waves) · GPA 4.6/5.0

Thesis: Magnetoelastic waves in irregular structures

Advisor: Evgeny Khorov (Director of IRE RAS)

Russian Government Scholarship

#### M.Sc. Core Courses

**Signal Processing & Communications** — Adaptive Filtering Theory · Fundamentals of Digital Signal Processing · Digital Signal Processing Laboratory · Digital Transmission Systems with OFDM · Spatial Signal Processing in Modern Wireless Communications · DSP in Fiber-Optic Communication Lines

**Radio Electronics & SDR** — Software Defined Radios (SDR) · EMC of Radioelectronic Equipment · Radio Electronic Systems

#### B.Sc. Mathematics

**Mathematical Analysis** — Introduction to Analysis · Multidimensional Analysis, Integrals and Series · Multiple Integrals and Field Theory · Harmonic Analysis

**Other Mathematics** — Functions of a Complex Variable · Equations of Mathematical Physics · Linear Algebra · Analytic Geometry · Differential Equations · Computational Mathematics

**Probability & Random Processes** — Probability Theory · Random Processes

#### B.Sc. Physics

**Theoretical Physics** — Analytical Mechanics · Field Theory · Quantum Mechanics · Statistical Physics

**General Physics** — Mechanics · Electricity & Magnetism · Thermodynamics & Molecular Physics · Optics · Quantum Physics

**Other** — Solid State Physics · Micro- and Nanoelectronic Semiconductor Devices

## COMPETITIONS & AWARDS

### Huawei Cup Wireless Communication Algorithm Contest — 1st Prize 2025

Lightweight neural network design and implementation for SVD of large-scale MIMO channel matrices; technical lead of a 2-person team, owning model lightweighting and training methodology; 1st Prize in a three-round contest, ranked 2nd overall out of 100+ teams.

### HSE Ph.D. Quota Selection — Winner 2023

University-wide doctoral topic competition; research proposal reviewed and defended before experts, awarding a full state-funded Ph.D. place.

### Russian Opendoors Olympiad, Physics Track — Winner 2021

Advanced comprehensive physics examination covering classical mechanics, electromagnetism, thermodynamics and quantum physics; awarded a full state-funded M.Sc. place.

## PUBLICATIONS

#### Journal Articles

- [1] **T. Dai**, E. Khorov, Li-WiGR: A Lightweight Wi-Fi-based Gesture Recognition System with CSI Subcarrier Grouping. *Journal of Communications Technology and Electronics* — accepted, to appear Dec. 2026.
- [2] **Т. Дай**, Е. Хоров, Li-WiGR: Легковесная система распознавания жестов на основе Wi-Fi с группированием поднесущих CSI. *Информационные процессы*, vol. 26, no. 2, 2026.
- [3] **T. Dai**, E. Khorov, WiZeCSI: Toward Wi-Fi-Based Single-Link Zero-Effort Cross-Domain Gesture Recognition. *IEEE Access* (Q1), 2026.
- [4] **T. Dai**, D.V. Kalyabin, S.A. Nikitov, Hypersonic magnetoelastic waves in inhomogeneous structures. *Ultrasonics* (Q1), 2022.

#### Conference Talks

- [1] **T. Dai**, E.M. Khorov, Wi-Fi-based few-shot open-set gesture recognition. **ICCT-2026** — 10th International Conference on Information, Control and Communication Technologies, Astrakhan, Russia.
- [2] **T. Dai**, E.M. Khorov, Система распознавания жестов на основе сигналов беспроводных сетей. **EN&T-2024** — International Conference on Engineering & Telecommunications, Dolgoprudny, Russia.